

QUESTIONS 20 / 20 completed

Cars Showroom

CARS SHOWROOM

Write a C program that uses a pointer to a structure to store information about a list of cars, including their make, model, year, and an array of prices for different conditions (e.g. "new", "used", "damaged"). Then, allow the user to enter a car's make and

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Car {
5     char make[20], model[20];
6     int year;
7     float price[3];
8 };
9
10 int main() {
11     struct Car c = {"Toyota", "Camry", 2024, {25000, 20000, 12000}};
12     struct Car *p = &c;
13     char make[20], model[20];
14
15     printf("Enter make and model: ");
16     scanf("%s %s", make, model);
17
18     if (strcmp(make, p->make) == 0 && strcmp(model, p->model) == 0) {
19         printf("New: %.2f\nUsed: %.2f\nDamaged: %.2f\n",
20             p->price[0], p->price[1], p->price[2]);
21     } else {
```

INPUT

Toyota Camry

QUESTIONS 20 / 20 completed

Book

BOOK

Define a structure in C for a book record that includes fields for the book's title, author, and an array of reviews. Each review should include the reviewer's name and a rating out of 5. Use Function.

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2 struct Review{char name[20];int rating;};
3 struct Book{char title[30],author[20];
4             struct Review r[3];};
5 void show(struct Book b){
6     printf("%s %s\n",b.title,b.author);
7     for(int i=0;i<3;i++)
8         printf("%s %d\n",b.r[i].name,b.r[i].rating);
9 }
10 int main()
11 {
12     struct Book b={"C Programming","Dennis",{"A",5},{"B",4},{"C",5}};
13     show(b);
14     return 0;
15 }
```

QUESTIONS 20 / 20 completed

Publication

PUBLICATION

Write a C program that uses an array of structures to store information about a list of books, including their titles, authors, and publication dates, and then sorts and prints out the books in order of publication date. Use Function.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Book{char title[30],author[30];int year;};
5
6 void sort(struct Book b[],int n){
7     struct Book t;
8     for(int i=0;i<n-1;i++)
9         for(int j=i+1;j<n;j++)
10             if(b[i].year>b[j].year)
11                 t=b[i],b[i]=b[j],b[j]=t;
12 }
13
14 int main(){
15     struct Book b[3]={{"Book3","Author3",2020},{"Book1","Author1",2010},{"Book2","Author2",2015}};
16     sort(b,3);
17     for(int i=0;i<3;i++) printf("%s %s %d\n",b[i].title,b[i].author,b[i].year);
18     return 0;
19 }
```

QUESTIONS 20 / 20 completed

Student

STUDENT

Define a structure in C for a student record that includes fields for the student's name, ID number, an array of grades, and the number of grades. Then, write a C program that calculates and prints out the student's average grade. Use Function.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 struct student
3 {
4     char name[50];
5     int id;
6     float grade[50];
7     int num_grade;
8 };
9 void std_rec(struct student *s)
10 {
11     float avg=0.0,sum=0.0;
12     int n=s->num_grade;
13     for(int i=0;i<n;i++)
14         sum+=s->grade[i];
15     avg=sum/n;
16     printf("std_avg grade=%.2f\n",avg);
17     printf("name=%s\n",s->name);
18     printf("id=%d",s->id);
19 }
20 int main()
21 {
```

QUESTIONS 20 / 20 completed

Grade

GRADE

Write a C program that uses an array of structures to store information about a list of cars, including their make, model, year, and an array of prices for different conditions (e.g. "new", "used", "damaged"). Then, allow the user to enter a car's make and

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Car {
5     char make[20];
6     char model[20];
7     int year;
8     float price[3];
9 };
10
11 void display(struct Car c) {
12     printf("New: %.2f\n", c.price[0]);
13     printf("Used: %.2f\n", c.price[1]);
14     printf("Damaged: %.2f\n", c.price[2]);
15 }
16
17 int main() {
18     struct Car cars[3] = {
19         {"Toyota", "Camry", 2024, {2500000, 1800000, 1200000}},
20         {"Honda", "City", 2023, {1800000, 1300000, 900000}},
21         {"Ford", "Figo", 2022, {1200000, 800000, 500000}}
```

QUESTIONS 20 / 20 completed

Earnings

EARNINGS

Define a structure in C for a company's quarterly earnings report that includes fields for the quarter (Q1, Q2, Q3, or Q4) and an array of earnings for each department. Then, write a C program that calculates and prints out the total earnings

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Earnings {
4     char quarter[3];
5     float earnings[4];
6 };
7
8 float total(struct Earnings e) {
9     float sum = 0;
10    for (int i = 0; i < 4; i++)
11        sum += e.earnings[i];
12    return sum;
13 }
14
15 int main() {
16     struct Earnings e[4] = {
17         {"Q1", {10000, 15000, 12000, 13000}},
18         {"Q2", {11000, 16000, 14000, 15000}},
19         {"Q3", {12000, 17000, 15000, 16000}},
20         {"Q4", {13000, 18000, 16000, 17000}}
21    };
```

QUESTIONS 20 / 20 completed

Rating



RATING

Write a C program that uses an array of structures to store information about a list of movies, including their titles, directors, actors, and an array of ratings out of 10 from different reviewers. Then, allow the user to enter a movie title and display

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Movie {
5     char title[30];
6     char director[30];
7     char actor[30];
8     float rating[3];
9 };
10
11 float average(struct Movie m) {
12     float sum = 0;
13     for (int i = 0; i < 3; i++)
14         sum += m.rating[i];
15     return sum / 3;
16 }
17
18 int main() {
19     struct Movie movies[3] = {
20         {"Leo", "Lokesh", "Vijay", {8, 9, 8}},
21         {"Jailer", "Nelson", "Rajini", {9, 8, 9}},
```

QUESTIONS 20 / 20 completed

Precipitation

PRECIPITATION

Define a structure in C for a weather report that includes fields for the temperature, humidity, wind speed, and an array of precipitation amounts for each day of the week. Then, write a C program that calculates and prints out the average

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Weather {
4     float temperature;
5     float humidity;
6     float windSpeed;
7     float precipitation[7];
8 };
9
10 float average(struct Weather w) {
11     float sum = 0;
12     for (int i = 0; i < 7; i++)
13         sum += w.precipitation[i];
14     return sum / 7;
15 }
16
17 int main() {
18     struct Weather w = {
19         30.5, 70.0, 12.5,
20         {10, 12, 8, 15, 5, 10, 12}
21     };
```

QUESTIONS 20 / 20 completed

Revenue

REVENUE

Write a C program that uses an array of structures to store information about a list of products, including their names, prices, and an array of quantities sold for each month. Then, calculate and print out the total revenue for each product.

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2
3 struct Product {
4     char name[20];
5     float price;
6     int quantity[3];
7 };
8
9 float revenue(struct Product p) {
10     int i, total = 0;
11     for (i = 0; i < 3; i++)
12         total += p.quantity[i];
13     return p.price * total;
14 }
15
16 int main() {
17     struct Product p[3] = {
18         {"Laptop", 50000, {2, 3, 1}},
19         {"Phone", 20000, {4, 5, 3}},
20         {"Tablet", 15000, {2, 2, 4}}
21     };
```

QUESTIONS 20 / 20 completed

Deposits

DEPOSITS

Define a structure in C for a bank customer's transaction record that includes fields for the transaction type (deposit or withdrawal), amount, date, and an array of descriptions. Then, write a C program that prints out the descriptions for all

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Transaction {
5     char type[15];
6     float amount;
7     char date[15];
8     char description[50];
9 };
10
11 void display(struct Transaction t[], int n, char type[]) {
12     int i;
13     for (i = 0; i < n; i++) {
14         if (strcmp(t[i].type, type) == 0)
15             printf("%s\n", t[i].description);
16     }
17 }
18
19 int main() {
20     struct Transaction t[4] = {
21         {"deposit", 5000, "01-08-2026", "Salary deposit"},
```

QUESTIONS 20 / 20 completed

Employee

EMPLOYEE

Write a C program that uses an array of structures to store information about a list of employees, including their names, salaries, and an array of bonuses for each quarter of the year. Then, calculate and print out the total earnings for each

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Employee {
4     char name[30];
5     float salary;
6     float bonus[4];
7 };
8
9 float totalEarnings(struct Employee e) {
10     float total = e.salary;
11     for (int i = 0; i < 4; i++)
12         total += e.bonus[i];
13     return total;
14 }
15
16 int main() {
17     struct Employee e[3] = {
18         {"Arun", 30000, {2000, 2500, 3000, 3500}},
19         {"Priya", 35000, {2500, 3000, 3500, 4000}},
20         {"Ravi", 40000, {3000, 3500, 4000, 4500}}
21     };
```

QUESTIONS 20 / 20 completed

Person Info

PERSON INFO

Define a structure in C for a person that includes fields for their name, age, and address. Then, write a function that takes a pointer to a person as an argument and prints out their name and age.

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2
3 struct Person {
4     char name[30];
5     int age;
6     char address[50];
7 };
8
9 void display(struct Person *p) {
10     printf("Name: %s\n", p->name);
11     printf("Age: %d\n", p->age);
12 }
13
14 int main() {
15     struct Person p = {"Varshini", 18, "Chennai"};
16
17     display(&p);
18
19     return 0;
20 }
```

QUESTIONS 20 / 20 completed

Modify

MODIFY

Write a C program that uses a pointer to a structure to modify the fields of a person record, including their name, age, and address.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Person {
5     char name[50];
6     int age;
7     char address[100];
8 };
9
10 int main() {
11     struct Person p;
12     struct Person *ptr = &p;
13
14     scanf("%99s", ptr->name);
15     scanf("%99d", &ptr->age);
16     scanf("%99s", ptr->address);
17
18     scanf("%99s", ptr->name);
19     scanf("%99d", &ptr->age);
20     scanf("%99s", ptr->address);
21 }
```

QUESTIONS 20 / 20 completed

Student Grade ▾

STUDENT GRADE

Define a structure in C for a student record that includes fields for the student's name, ID number, and an array of grades. Then, write a function that takes a pointer to a student record as an argument and calculates their average grade.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include <stdio.h>
2
3 struct Student {
4     char name[50];
5     int id;
6     float grades[5];
7 };
8
9 float average(struct Student *s) {
10     float sum = 0;
11     for (int i = 0; i < 5; i++)
12         sum += s->grades[i];
13     return sum / 5;
14 }
15
16 int main() {
17     struct Student s;
18
19     scanf("%99s", s.name);
20     scanf("%99d", &s.id);
21
```

INPUT

Varshini
101
80 90 85 95 90

QUESTIONS 20 / 20 completed

Max-Min

MAX-MIN

Write a C program that uses a pointer to a structure to find the maximum and minimum values in an array of integers.

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2
3 struct Result {
4     int max;
5     int min;
6 };
7
8 int main() {
9     struct Result r;
10    int a[5], i;
11    int *p = a;
12
13    for (i = 0; i < 5; i++)
14        scanf("%d", p + i);
15
16    r.max = r.min = *p;
17
18    for (i = 1; i < 5; i++) {
19        if (*(p + i) > r.max)
20            r.max = *(p + i);
21        if (*(p + i) < r.min)
```

INPUT

10 25 5 40 15

QUESTIONS 20 / 20 completed

Bank

BANK

Define a structure in C for a bank customer's transaction record that includes fields for the transaction type (deposit or withdrawal), amount, date, and an array of descriptions. Then, write a function that takes a pointer to a transaction record as

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Transaction {
5     char type[10];
6     float amount;
7     char date[15];
8     char desc[3][50];
9 };
10
11 void show(struct Transaction *t, int n, char *type) {
12     int i, j;
13     for (i = 0; i < n; i++) {
14         if (strcmp(t[i].type, type) == 0) {
15             printf("\nType: %s", t[i].type);
16             printf("\nAmount: %.2f", t[i].amount);
17             printf("\nDate: %s", t[i].date);
18             printf("\nDescriptions:");
19             for (j = 0; j < 3; j++)
20                 printf("\n%s", t[i].desc[j]);
21         }
```

QUESTIONS 20 / 20 completed

Products

PRODUCTS

Write a C program that uses a pointer to a structure to store information about a list of products, including their names, prices, and quantities sold. Then, calculate and print out the total revenue for each product.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Product {
4     char name[30];
5     float price;
6     int quantity;
7 };
8
9 int main() {
10     struct Product p[3] = {
11         {"Laptop", 50000, 2},
12         {"Mobile", 20000, 3},
13         {"Headphones", 2000, 5}
14     };
15
16     struct Product *ptr = p;
17     int i;
18
19     for (i = 0; i < 3; i++) {
20         printf("Product: %s\n", ptr[i].name);
21         printf("Price: %.2f\n", ptr[i].price);
```

QUESTIONS 20 / 20 completed

Earnings

EARNINGS

Define a structure in C for a company's quarterly earnings report that includes fields for the quarter (Q1, Q2, Q3, or Q4) and an array of earnings for each department. Then, write a function that takes a pointer to an earnings report as an

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Earnings {
4     char quarter[3];
5     float e[5];
6 };
7
8 float total(struct Earnings *p) {
9     float sum = 0;
10    for (int i = 0; i < 5; i++)
11        sum += p->e[i];
12    return sum;
13 }
14
15 int main() {
16    struct Earnings x;
17
18    scanf("%99s", x.quarter);
19    for (int i = 0; i < 5; i++)
20        scanf("%99f", &x.e[i]);
21
```

INPUT

Q1
1000
2000
1500
2500
3000

QUESTIONS 20 / 20 completed

Weather ▾

WEATHER

Write a C program that uses a pointer to a structure to modify the fields of a weather report, including the temperature, humidity, wind speed, and precipitation amounts for each day of the week.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include <stdio.h>
2
3 struct Weather {
4     float temp, humidity, wind, rain;
5 };
6
7 int main() {
8     struct Weather w[7], *p = w;
9     int i;
10
11     for(i = 0; i < 7; i++) {
12         scanf("%f%f%f%f", &p[i].temp, &p[i].humidity,
13             &p[i].wind, &p[i].rain);
14
15         p[i].temp += 1;
16         p[i].humidity += 2;
17         p[i].wind += 1;
18         p[i].rain += 0.5;
19     }
20
21     for(i = 0; i < 7; i++)
```

INPUT

```
25 60 10 2
26 65 12 1
24 55 8 3
27 70 15 0
23 50 9 2
28 75 14 1
22 45 7 4
```

QUESTIONS 20 / 20 completed

Book Publication ▾

BOOK PUBLICATION

Define a structure in C for a book record that includes fields for the book's title, author, and an array of reviews. Each review should include the reviewer's name and a rating out of 5. Then, write a function that takes a pointer to a book record as an

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include <stdio.h>
2
3 struct Review {
4     char name[20];
5     int rating;
6 };
7
8 struct Book {
9     char title[50];
10    char author[50];
11    struct Review r[5];
12    int n;
13 };
14
15 float average(struct Book *b) {
16     int sum = 0;
17     for (int i = 0; i < b->n; i++)
18         sum += b->r[i].rating;
19
20     return (float)sum / b->n;
21 }
```

INPUT

C Programming
Dennis Ritchie
3
Ravi
4
Priya
5
Arun
3